# Introduction to Statistics

Kaung Myat Khant

# Statistics

## 1 Data and **variables**
Statistics & probability
R setup and GUI

## 2 Data tidying
Tidy data concept
data tidying

## 3 Data summarizing
Descriptive statistics
Exploratory data analysis

## 4 Inference
Probability distributions
Estimation

## 5 Hypothesis testing
Z test, T test
One way ANOVA
Chi square and Fisher exact

## 6 Regression analysis
Linear regression
Logistic regression

There are two things extremely difficult to understand:

- Statistics
- Women

P value <0.001

- **မတူဘူးဆိုတာက တကယ်မဟုတ်ဘူး ဆိုတာကို မှားယွင်းငြင်းဆိုမိဖို့ ဖြစ်နိုင်ခြေ တစ်ထောင်ပုံတစ်ပုံရှိတယ်။**

- **မှားယွင်းဖို့ ဖြစ်နိုင်ခြေ မရှိသလောက်နည်းတယ်။**

# A Time-Series Analysis of my Girlfriends Mood Swings

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**Abstract**

Despite recent advances in active listening, date night, and extended pillow talk, it is becoming increasingly more difficult to forecast Tiffany's mood. With more and more Playstation 5 exclusive games, it is becoming increasingly important to determine Tiffany's mood before purchasing a new game and playing online Co-op with the boys every evening for a week straight. This paper aims to determine the optimal forecast model of my girlfriend's drastically growing mood swings by comparing simple moving averages, to sextuple exponential smoothing and even an overly complicated Machine Learning model. Despite initial analysis showing non-stationarity and highly seasonal mood swings, the more simple models provided less riskier forecast predictions when planning a three day bender after Matt got divorced.

Keywords: Relationships, Time Series Analysis, Forecast Modeling, Playstation 5 Exclusives

## 1. Introduction

Traditional methods of determining whether Tiffany is in a good mood has produced wildly subjective results causing dangerous outcomes such as her taking three hours to respond to a text, flaking on netflix and chill plans or even most disastrously having to return a speed boat even though it was a great deal and an even better investment.

Just asking her if she's okay is not enough anymore. The only reliable method is to develop, test, verify, and implement an extensive forecasting model by analyzing historical Tiffany mood swing data.

### *1.1 Background*

Tiffany and I met at Cranberry-Lemon University as sophomores in our BS required Theoretical Physical Education class ten years ago. After being the last two in the quantum particle dodgeball match, we began an on again, off again relationship until our Junior fall semester when it became too cold to go outside. She and I are two young professionals living in our own home which we financed using a downpayment from not eating avocado toast for three years while I finished my doctorate program.

Tiffany is now a freemium gaming marketing consultant who hates it when I call it freemium gaming. She loves Disney, prefers beach vacations over mountains, and was obsessed with Game of Thrones until the end of the last season. When she's not binge watching the office, you can find her endlessly scrolling through reddit for memes, facebook or instagram for jealousy, and twitter to keep up on all the public officials and celebrities she hates the most.

### *1.2 Purpose*

Ever since her best friends started having kids and she got promoted to a stressful corporate position she was not trained for, it has become exponentially difficult to plan around Tiffany's emotional highs and lows. This is becoming not only problematic but is a problem that needs to be solved immediately. To dispel any misconceptions, it's not about avoiding her negative mood swings with asinine boyfriend behavior but also taking advantage of her positive mood swings for the least risky time to hang out with the boys or even more risky, refinance the house so I can get that boat back. Most importantly, there is one known application that has expedited the development of this forecast model. The Final Fantasy 7 remake will be released soon this June in under a month!

The amount of time it is expected to take to 100% complete the game, despite playing the original many times, must be carefully scheduled around a reliable Tiffany Mood Forecast Model (TMFM). Being able to plan around her mood will not only allow enough time to max out Cloud's stats but create more opportunities for other future video game releases while keeping Tiffany happy.

Initial analysis of historical screen time and purchase history data has shown that Tiffany's mood is not only

# What do we do?

- Must
  - Attend class regularly
  - n’t be shy to ask
  - Never learn alone
  - Upgrade your computer literacy
- Should do
  - Read reference and take your own notes
  - Watch YouTube
- Good to do
  - Literature review and take notes
  - Discuss often with friends

# What is statistics?

- စာရင်းအင်း သင်္ချာ
- စာရင်း
- အင်း
- သင်္ချာ

ဝါဆို
အာလူး - ၁၀သား
ကြက်သွန် - ၂၀ သား

ဝါခေါင်
အာလူး - ၂၀သား
ကြက်သွန် - ၃၀ သား

အာလူး - ၁၀သား ၃၀၀
ကြက်သွန် - ၁၀ သား ၅၀၀

| လ | စားကုန် | ကျပ်သား | ကျသင့်ငွေ |
|---|---|---|---|
| ဝါဆို | အာလူး | ၁၀ | ၃၀၀ |
| ဝါဆို | ကြက်သွန် | ၂၀ | ၁၀၀၀ |
| ဝါခေါင် | အာလူး | ၂၀ | ၆၀၀ |
| ဝါခေါင် | ကြက်သွန် | ၃၀ | ၁၅၀၀ |

| | ဝါဆို | ဝါခေါင် | နှစ်လပေါင်း |
|---|---|---|---|
| အာလူး | ၃၀၀ | ၆၀၀ | ၉၀၀ |
| ကြက်သွန် | ၁၀၀၀ | ၁၅၀၀ | ၂၅၀၀ |
| နှစ်ခုပေါင်း | ၁၃၀၀ | ၂၁၀၀ | ၃၄၀၀ |

+ − × ÷

# Why statistics?

- Smart TV ၄၂လက်မ – ၇၀၀,၀၀၀
- လစာ - ၂၅၀,၀၀၀
- ၇လကြာလျှင် တီဗီဝယ်နိုင်ရန် တရက်မည်မျှသုံးရမည်နည်း။
- တီဗီစျေး ခုနစ်လအတွင်း မပြောင်းလဲ
- ဆေးရုံတက် ဆိုင်ကယ်ပျက်ခြင်း မဂ်လာဆောင်လက်ဖွဲ့ ခုနစ်လအတွင်း မရှိ

# Why statistics?

Statistics = State craft

ကျနော်တို့နိုင်ငံမှာ ကျောင်းနေအရွယ် ကလေး
ဘယ်လောက်ရှိလဲ။ ကလေးတိုင်းပညာသ
ဘတ်ဂျက်ဘယ်လောက်သုံးရမလဲ

ကျနော်တို့နိုင်ငံမှာ မွေးဖွားနှုန်း ၂.၁ ရှိရင်
နောက်ဘဝနှစ်မှာ ကုန်ထုတ်လုပ်နိုင်စွမ်း
ဘယ်လောက်တက်လာမလဲ

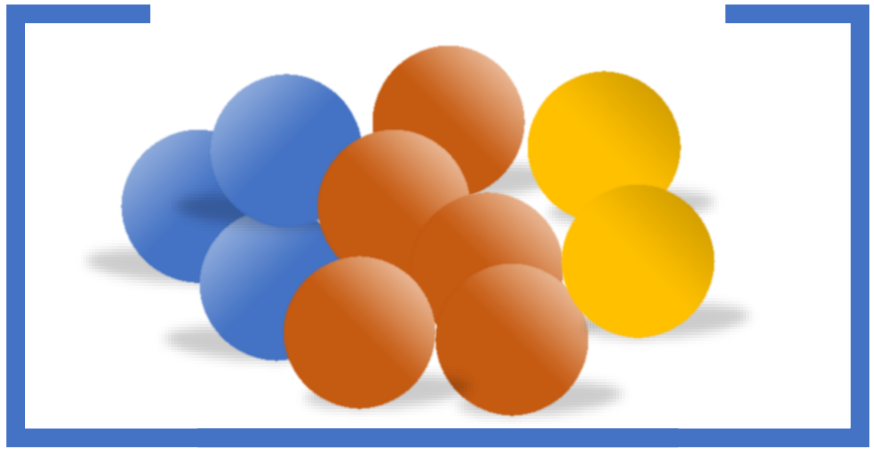
**Uncertainty**

အတွဲတွေပြတ်ရင် အမျိုးသား
ဘယ်သူက စိတ်ကျရောဂါ
ပိုများသလဲ။

ကာကွယ် ထိန်းချုပ်ရေး နည်းလမ်းတွေကို
ိုင်အထည် ဖော်နိုင်ခဲ့ရင် ခုတင်နဲ့ သူနာပြု ဘယ်လောက်
လိုမလဲ။ မထိန်းချုပ်နိုင်ခဲ့ရင် ခုတင် နဲ့ သူနာပြု
ဘယ်လောက်လိုမလဲ။

# PROBABILITY

A proportion of times a particular event would occur in a large number of similar repeated trial



- Expressed as percentage
  - Range from 0-100%
  - 0 – never occur
  - 100% - certain to occur
  - Event denoted with a capital letter
    - P(A), P(B)

# PROBABILITY

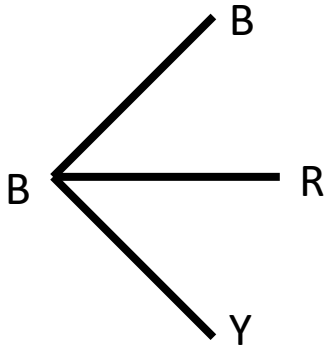
## Class work

- ဘောလုံးတစ်လုံးကို ကျပန်းနှိုက်မယ်။ အရောင်မှတ်မယ်။ ပြန်ထည့်မယ်။နောက်တစ်ခါ ကျပန်းပြန်နှိုက်မယ်။ နှစ်ခါလုံး အပြာဖြစ်ဖို့ ဖြစ်တန်စွမ်း ဘယ်လောက်ရှိလဲ။
- ဘောလုံးတစ်လုံးကို ကျပန်းနှိုက်မယ်။ အရောင်မှတ်မယ်။ ပြန်မထည့်ဘူး။နောက်တစ်ခါ ကျပန်းပြန်နှိုက်မယ်။ နှစ်ခါလုံး အပြာဖြစ်ဖို့ ဖြစ်တန်စွမ်း ဘယ်လောက်ရှိလဲ။

Conditional probability

Independent

Dependent



# Probability and Expected Value

အံစာတုံးကျတဲ့ ဂဏန်းကို ၁၀၀နဲ့မြှောက်ပြီး ဒေါ်လာနဲ့ပေးမယ်။

သင် ဘယ်လောက်ရမယ် ခန့်မှန်းလဲ။ ခန့်မှန်းတာ မှားခဲ့ရင် အမှန်နဲ့ကွာသလောက်ပြန်လျော်ရမယ်။

100, 200,300,400,500,600

$$Expected\ value = (1 \quad \times 100) \times \frac{1}{6} + (2 \quad \times 100) \times \frac{1}{6} + (3 \quad \times 100) \times \frac{1}{6} + (4 \quad \times 100) \times \frac{1}{6} + (5 \quad \times 100) \times \frac{1}{6} + (6 \quad \times 100) \times \frac{1}{6}$$

Mean

$$\boxed{\frac{(100 + 200 + 300 + 400 + 500 + 600)}{6} = 350}$$

$$\frac{300 + 400}{2} = 350$$

Median

| | 350 |
|---|---|
| 100 | 250 |
| 200 | 150 |
| 300 | 50 |
| 400 | 50 |
| 500 | 150 |
| 600 | 250 |

# Probability and Expected Value

အံစာတုံးကျတဲ့ ဂဏန်းရဲ့နှစ်ထပ်ကိန်းကို ၁၀၀နဲ့မြှောက်ပြီး ဒေါ်လာနဲ့ပေးမယ်။

သင် ဘယ်လောက်ရမယ် ခန့်မှန်းလဲ။ ခန့်မှန်းတာ မှားခဲ့ရင် အမှန်နဲ့ကွာသလောက်ပြန်လျော်ရမယ်။

100, 400,900,1600,2500,3600

$$Expected\ value = (1^2 \times 100) \times \frac{1}{6} + (2^2 \times 100) \times \frac{1}{6} + (3^2 \times 100) \times \frac{1}{6} + (4^2 \times 100) \times \frac{1}{6} + (5^2 \times 100) \times \frac{1}{6} + (6^2 \times 100) \times \frac{1}{6}$$

$$\frac{(100 + 400 + 900 + 1600 + 2500 + 3600)}{6} = 1533.33$$

$$\frac{900 + 1600}{2} = 1250$$

| | | |
|---|---|---|
| | 1533.33 | 1250 |
| 100 | 1433.33 | 1150 |
| 400 | 1133.33 | 850 |
| 900 | 633.33 | 350 |
| 1600 | 66.67 | 350 |
| 2500 | 966.67 | 1250 |
| 3600 | 2066.67 | 2350 |

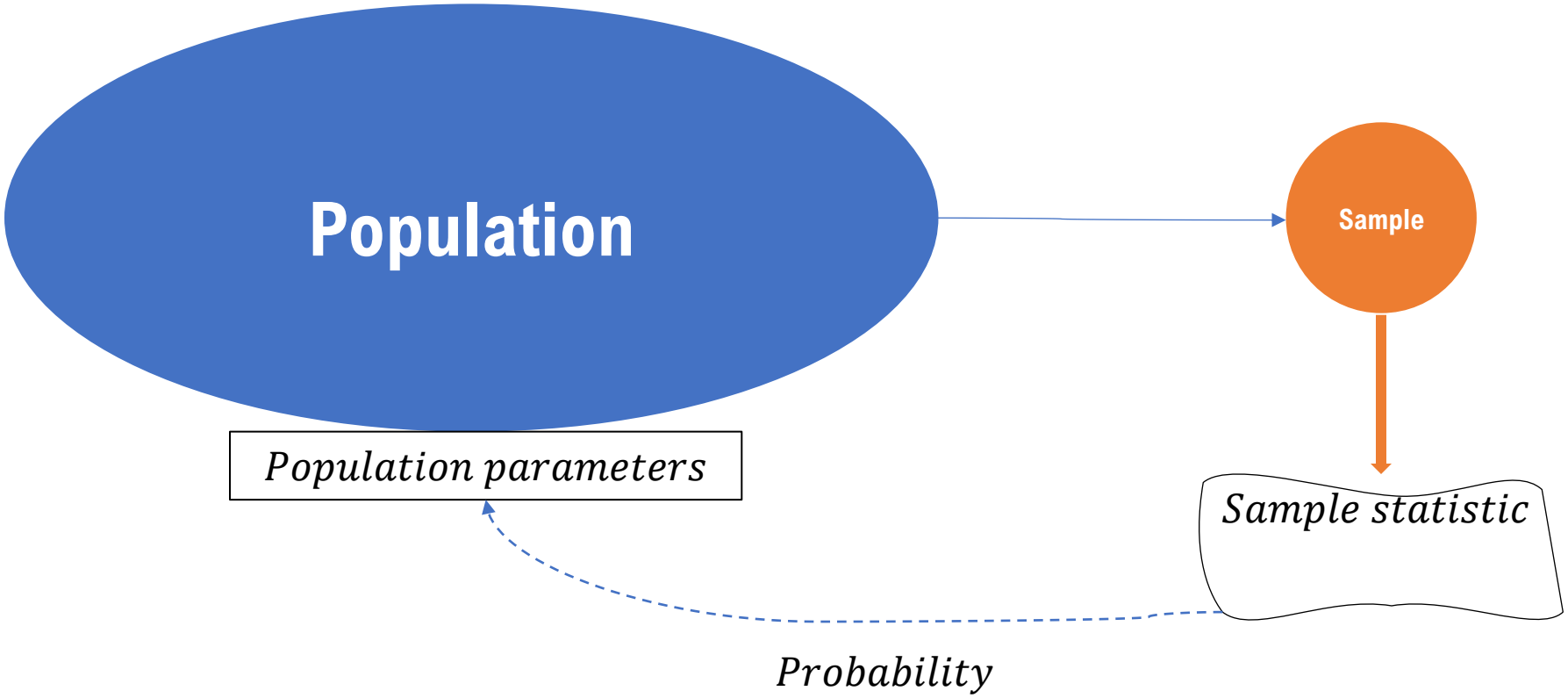
# Why statistics?

- Expected values calculated using probability
- Expected values are measures of central tendency
- Expected values are used for:
  - Decision making
  - Hypothesis testing
  - Predicting probability

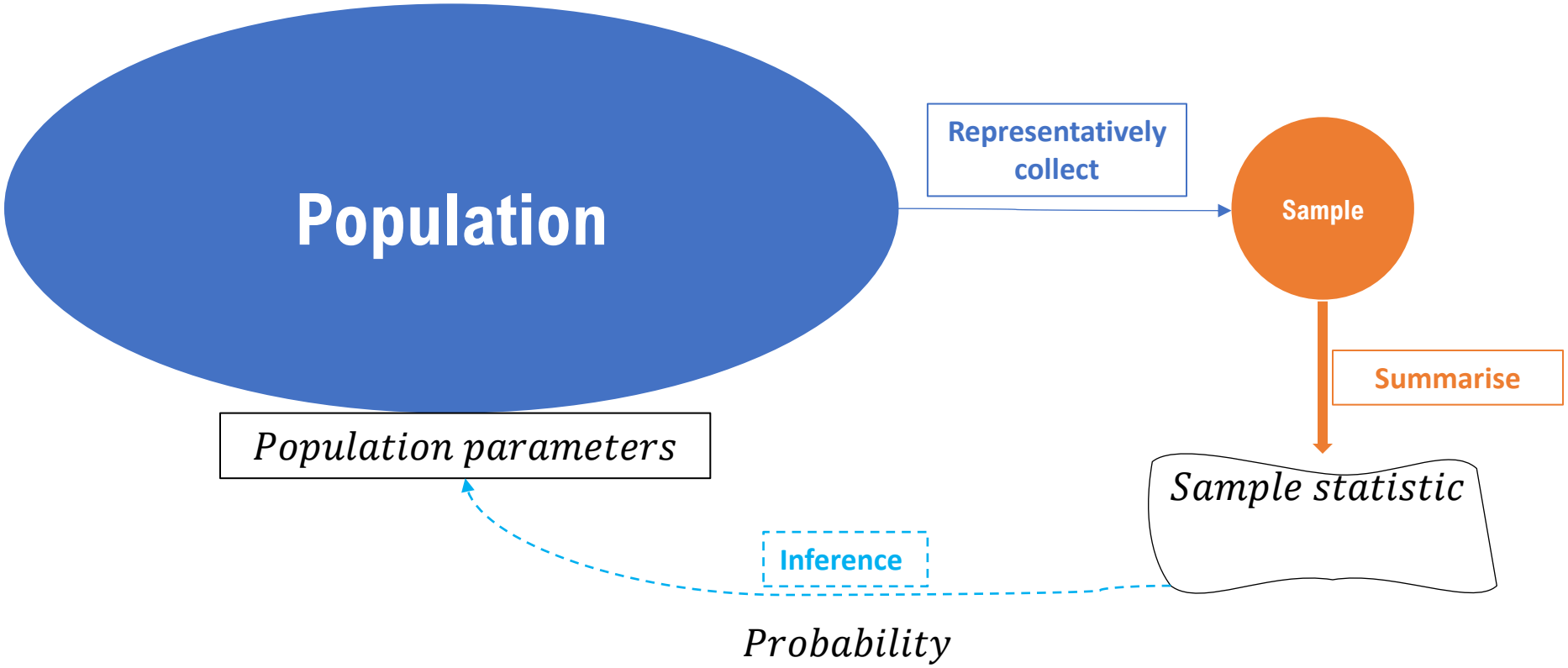
# Why not statistics only?

- Story behind the data is also important
- Statistics depends on assumptions
- The world is dynamic/chaos
- The data may be incomplete or biased

# Statistics


Population
Sample
Population parameters
Sample statistic
Probability

# Statistics



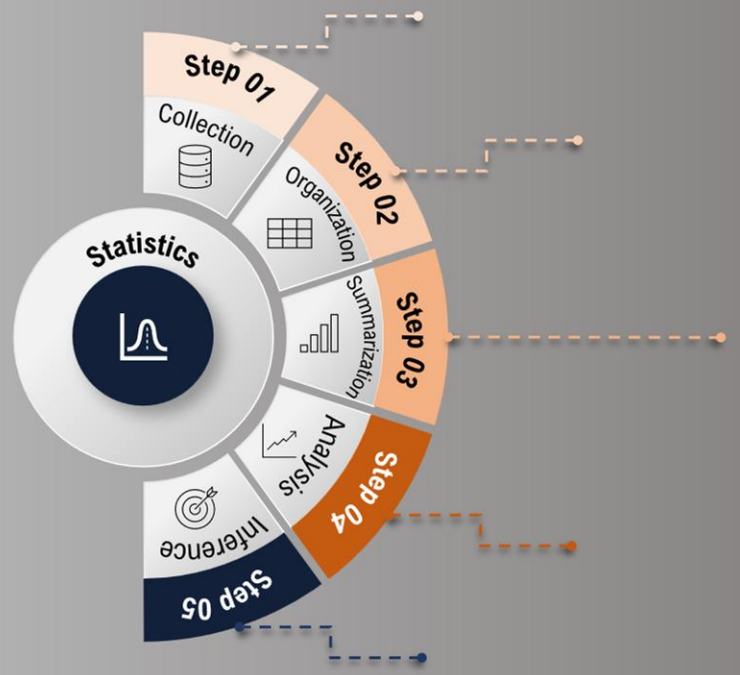

**Idea credit to Dr. Kyaw Swa Mya**

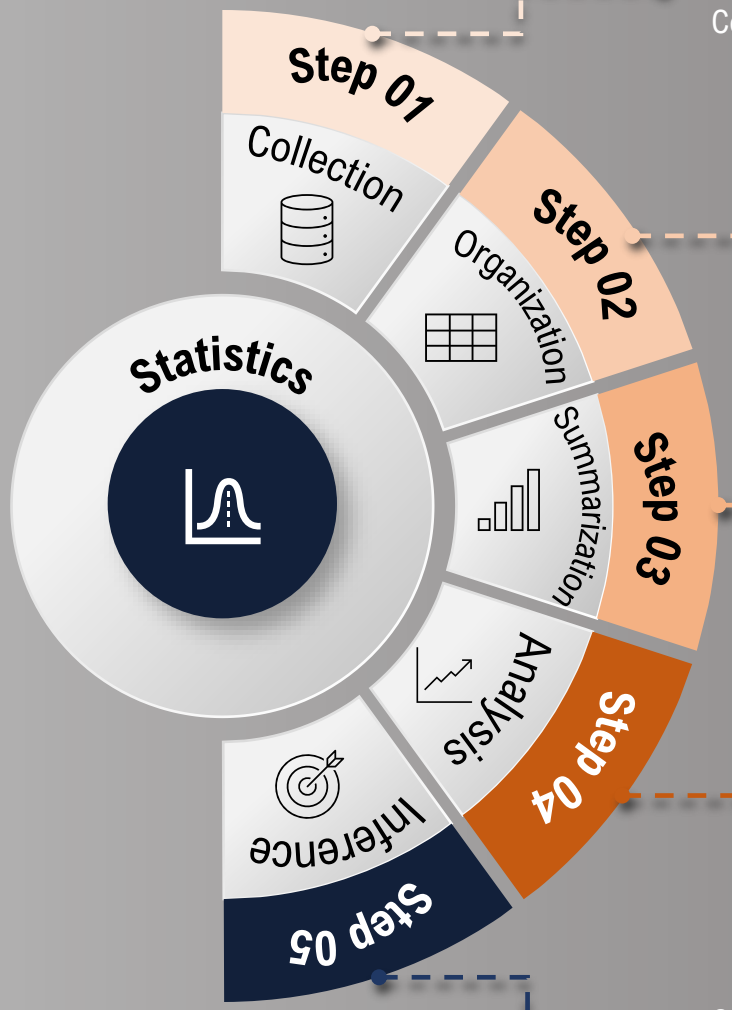
# Definition of statistics

A field of study concerned with

- Collection
- Organization
- Summarization
- Analysis
- Drawing of inference about population

sample






## 01. Data Collection

Select representative sample
Collect information about the sample

## 02. Data Organization

Organize the collected data in appropriate rows and columns
Tidy the dataset according to analysis purpose

## 03. Data Summarization

Summarize  data using appropriate summary measures
Bar graph, Pie chart, Dot Plots, Histogram, Boxplot
Measure of central tendencies and dispersion

## 04. Data analysis

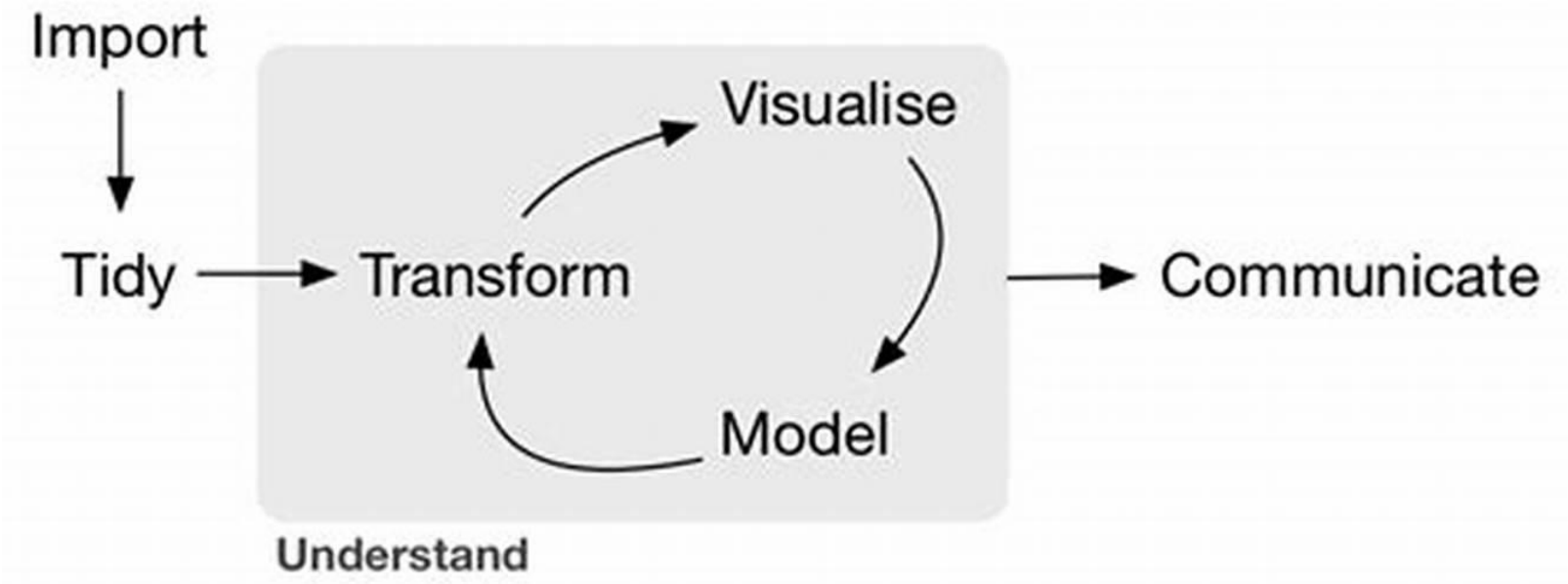
Compare the summarized data  depending on subgroup
Meaningful insight of data
Patterns and trends
Relationship and correlation

## 05. Population inference

Use data analysis to make conclusion about the population based on sample data
Estimate population parameters
Test hypothesis
Make prediction

# Modern-day Statistics

- BI, Data science, Machine learning, Deep learning, AI, Neural Network



# Thanks for your attention